

ORIGINAL ARTICLE

Identification of the histidine-rich glycoprotein domains responsible for contact pathway inhibition

Tammy K. Truong^{1,2,3}  | Rida A. Malik^{1,3}  | Xintong Yao^{1,3}  |
James C. Fredenburgh^{1,2}  | Alan R. Stafford^{1,2} | Hasam M. Madarati^{1,3}  |
Colin A. Kretz^{1,2}  | Jeffrey I. Weitz^{1,2,3,4} 

¹Thrombosis and Atherosclerosis Research Institute, Hamilton, Ontario, Canada

²Department of Medicine, McMaster University, Hamilton, Ontario, Canada

³Department of Medical Sciences, McMaster University, Hamilton, Ontario, Canada

⁴Department of Biochemistry and Biomedical Sciences, McMaster University, Hamilton, Ontario, Canada

Correspondence

Jeffrey I. Weitz, Thrombosis and Atherosclerosis Research Institute, 237 Barton St. E., Hamilton, Ontario, Canada L8L 2X2.
Email: weitzj@taari.ca

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Abstract

Background: Previously, we showed that histidine-rich glycoprotein (HRG) binds factor (F) XIIa with high affinity, inhibits FXII autoactivation and FXIIa-mediated activation of FXI, and attenuates ferric chloride-induced arterial thrombosis in mice. Therefore, HRG downregulates the contact pathway *in vitro* and *in vivo*.

Objective: To identify the domains on HRG responsible for contact pathway inhibition.

Methods: Recombinant HRG domain constructs (N-terminal [N1, N2, and N1N2], proline-rich regions, histidine-rich region [HRR], and C-terminal) were expressed and purified. The affinities of plasma-derived HRG, HRG domain constructs, and synthetic HRR peptides for FXII, FXIIa, β -FXIIa, and polyphosphate (polyP) were determined using surface plasmon resonance, and their effects on polyP-induced FXII autoactivation, FXIIa-mediated activation of FXI and prekallikrein, the activated partial thromboplastin time (APTT), and thrombin generation were examined.

Results: HRG and HRG domain constructs bind FXIIa, but not FXII or β -FXII. HRR, N1, and N1N2 bind FXIIa with affinities comparable with that of HRG, whereas the remaining domains bind with lower affinity. Synthetic HRR peptides bind FXIIa and polyP with high affinity. HRG and HRR significantly inhibit FXII autoactivation and prolong the APTT. Like HRG, synthetic HRR peptides inhibit FXII autoactivation, attenuate FXIIa-mediated activation of prekallikrein and FXI, prolong the APTT, and attenuate thrombin generation.

Conclusion: The interaction of HRG with FXIIa and polyP is predominantly mediated by the HRR domain. Like intact HRG, HRR downregulates the contact pathway and contributes to HRG-mediated down regulation of coagulation.

KEYWORDS

coagulation, factor XII, histidine-rich glycoprotein, polyphosphate, thrombosis

1 | INTRODUCTION

The contact pathway is triggered when factor (F) XII binds to negatively charged surfaces and undergoes autoactivation.^{1,2} Such surfaces include nucleic acids released from cells undergoing apoptosis, autophagy, or necrosis, neutrophil extracellular traps extruded from activated neutrophils, and short chain polyphosphate (polyP) consisting of 70–100 phosphate units released from activated platelets.^{3–7}

Binding of FXII to negatively charged surfaces induces its autoactivation via a conformational change that is enhanced in the presence of Zn^{2+} .^{8,9} FXII autoactivation occurs by cleavage of the Arg353-Val354 bond, which converts single-chain FXII into its activated, two-chain form.^{10–12} α -FXIIa (FXIIa) propagates coagulation by activating FXI, which circulates in complex with high molecular weight kininogen (HK), and culminates in thrombin generation and fibrin formation.^{13,14} Recent studies suggest that FXII, which is dispensable for hemostasis, is important for thrombus growth.^{15–23} Therefore, FXIIa inhibition may attenuate thrombosis without disrupting hemostasis.

Previously, we showed that histidine-rich glycoprotein (HRG) binds FXIIa with high affinity, thereby inhibiting FXII autoactivation and FXIIa-mediated activation of FXI,²⁴ and attenuating ferric chloride ($FeCl_3$)-induced arterial thrombosis in mice.²⁵ HRG is a 75-kDa glycoprotein that circulates in plasma at a concentration of ~1–3 μ M and is implicated in diverse processes including coagulation, fibrinolysis, immunity, and angiogenesis.^{24,26–28} Human HRG consists of two cystatin-like N-terminal domains (N1 and N2), a central histidine-rich region (HRR) flanked by two proline-rich regions (PRR1 and PRR2), and a C-terminal domain (COOH).^{29,30} The HRR is unique because of its unusually high content of histidine (13% of total amino acids) and the presence of 12 tandem repeats of a Gly-His-His-Pro-His consensus sequence.²⁹ The multidomain structure of HRG endows it with the capacity to bind many ligands including fibrinogen, plasminogen, thrombospondin, and heparin.^{26,27,31,32} The HRR domain of HRG binds divalent metal cations (e.g., Zn^{2+} , Ni^{2+} , Cu^{2+} , Co^{2+}) and undergoes a conformational change, which may influence its biological activity.^{26,27,33} HRG also binds nucleic acids and polyP and attenuates their capacity to activate the contact pathway.^{34,35} Therefore, HRG is a multifunctional protein.

HRG downregulates coagulation in at least two ways. First, HRG binds to an exosite on FXIIa and prevents its interactions with FXII and FXI.²⁴ HRG does not bind to zymogen FXII, FXI, FXIa, or FIX.²⁴ Second, HRG binds polyP and other polyanions and attenuates their capacity to activate FXII.^{24,34,35} Although HRG is a potent regulator of the intrinsic pathway of coagulation *in vitro*²⁴ and *in vivo*,²⁵ the HRG domains responsible for its interaction with FXIIa and polyanions such as polyP remain elusive. To identify these domains, we generated recombinant constructs of individual HRG domains and synthetic HRR peptide analogs and determined their (1) affinities for FXIIa, FXII, β -FXIIa, and polyP; (2) inhibitory effects on polyP-induced FXII autoactivation, FXIIa-mediated activation of FXI, and prekallikrein (PK); and (3) effects on the activated partial thromboplastin time (APTT) and polyP-induced thrombin generation.

ESSENTIALS

- Histidine-rich glycoprotein (HRG) downregulates the contact pathway.
- The histidine-rich region (HRR) domain of HRG modulates factor XIIa activity.
- HRR analogs recapitulate the effects of intact HRG.
- Like HRG, HRR analogs may attenuate thrombosis without disrupting hemostasis.

2 | METHODS

2.1 | Reagents

Oligonucleotide primers were from MOBIX (McMaster University, Hamilton, ON) and are described in Table S1. HRG (NM_000412) Human Tagged ORF Clone (cDNA) was purchased from OriGene Global (Rockville, MD). The expression vector, pMAL-c5e, and BL21-Gold (DE3) *Escherichia coli* were from New England Biolabs Ltd. (Whitby, ON). Peptide analogs of the HRR containing one or four repeats of the His-His-Pro-His-Gly (HHPHG)_{1–4} consensus sequence,³⁶ a scrambled peptide containing five repeats of HAHG, Scr(HAHG)₅H, and poly-His peptides consisting of six or 12 His residues (H-6 and H-12) were from Biomatik (Cambridge, ON) (Table S2). Short chain polyP consisting of 70–100 phosphate units (P70.5) was a generous gift from Dr. Thomas Stoffel (BK Giulini GmbH, Ladenburg, Germany). Biotinylated Phe-Pro-Arg-chloromethyl ketone was from Hemalogic Technologies Inc. (Essex Junction, VT). Zinc chloride ($ZnCl_2$), silicon dioxide (silica; 5–15 nm), IPTG, and EDTA were purchased from Sigma Aldrich (St. Louis, MO). Human FXII, α -FXIIa, β -FXIIa, FXI, PK, HK, and corn trypsin inhibitor (CTI) were from Enzyme Research Laboratories (South Bend, IN). Unless otherwise specified, FXIIa will refer to α -FXIIa. RecombiPlasTin 2G and APTT-SP (liquid) were purchased from Instrumentation Laboratory Company (Bedford, MA). Phosphatidylcholine-phosphatidylserine (3:1) vesicles were prepared as previously described.³⁷ A sheep polyclonal antibody against human HRG and a polyclonal goat anti-human FXII antibody were from Affinity Biologicals (Ancaster, ON). The HRG-directed antibody was affinity purified as previously described.²⁵ Horseradish peroxidase (HRP) conjugate was prepared using a Lightning-Link HRP conjugation kit from Novus Biologicals (Centennial, CO) according to the manufacturer's protocol. A polyclonal rabbit antibody against maltose-binding protein (MBP) antibody was from Thermo Fisher Scientific (Burlington, ON). BIAcore sensor chips (CM4 and CM5) and amine coupling kit were from Cytiva (Mississauga, ON). HRG was isolated from human plasma using a Ni-NTA agarose column and human plasma was depleted of HRG by affinity chromatography as previously described.²⁴

2.2 | Cloning, expression, and purification of MBP-HRG domain constructs

2.2.1 | Cloning

Primers were designed to polymerase chain reaction-amplify individual and adjacent domains of human HRG (Table S1). Polymerase chain reaction products were gel purified, digested with *Ava*I or *Nde*I and *Bam*H1 (New England Biolabs Ltd.), and then ligated into a pMAL-c5e expression vector containing a NH₂-terminal MBP-tag. The domain constructs were as follows: MBP-N1, MBP-N2, MBP-N1N2, MBP-PRR1, MBP-HRR, MBP-PRR2, and MBP-COOH (Figure S1). All clones were verified by Sanger sequencing.

2.2.2 | Expression

MBP-HRG domain plasmids were transformed into BL21 (DE3) *E coli* cells and allowed to grow for 1–2 h at 37°C until reaching mid-log phase ($A_{600} = \sim 0.6$). Expression was induced with 1 mM IPTG for an additional 4 h at 37°C. Cells were harvested by centrifugation at 4000g for 20 min and the supernatant was discarded. The pellet was resuspended in 25 ml of column buffer (20 mM Tris-HCl, 200 mM NaCl, 1 mM EDTA, 1 mM sodium azide, pH 7.4) in the presence of Complete Protease Inhibitor (Sigma). Bacteria were placed in an ice water bath and subjected to lysis by sonication in 5 s on/off pulses for 20 min using a FB120 Sonic Dismembrator fitted with a CL-18 probe (Fisher Scientific). Lysed bacteria were subjected to centrifugation at 20 000g for 20 min. The supernatant was harvested and diluted 1:6 with column buffer and subjected to amylose affinity chromatography at 4°C.

2.2.3 | Purification

Amylose resin (New England Biolabs) in a 10-ml column was washed with five volumes of column buffer. Diluted supernatant was loaded onto the column and the flow through was collected. The column was washed with 12 volumes of column buffer and protein was eluted with 15 ml of 10 mM maltose (New England Biolabs) in column buffer. One-milliliter fractions were collected, and protein content was assessed by measuring absorbance at 280 nm and 320 nm. Protein-containing fractions were pooled; subjected to buffer exchange four times into 20 mM HEPES-OH, 150 mM NaCl, pH 7.4 (HBS); concentrated using Amicon Ultra-15 Centrifugal Filter Units (Thermo Fisher Scientific, Inc.); and stored in aliquots at –80°C. Protein concentrations were determined by measuring absorbance and using calculated extinction coefficients (ExPASy ProtParam online tool).

2.3 | SDS-PAGE and Western blot analysis

Purified recombinant MBP-HRG domain constructs were subjected to SDS-PAGE analysis on 4%–15% polyacrylamide gels

(Bio-Rad Laboratories Ltd.) in the presence of 0.1% SDS and 0.14 M β -mercaptoethanol. Separated proteins were transferred onto nitrocellulose membranes (Bio-Rad) overnight at 4°C using a Mini-PROTEAN Tetra electrophoresis system (Bio-Rad). Membranes were blocked with 1% BSA in 50 mM Tris-HCl, 150 mM NaCl, pH 7.4 containing 0.005% Tween20 and probed with affinity-purified HRP-conjugated polyclonal sheep anti-HRG antibody (0.2 μ g/ml) for 2 h or affinity-purified polyclonal rabbit antibody against MBP (0.1 μ g/ml) overnight at 4°C. Blots probed with the antibody against MBP were then incubated with 0.2 μ g/ml HRP-linked secondary anti-rabbit antibody (Cedarlane) for 2 h at 25°C. Bound HRP was detected with the Immune-Star HRP kit (Bio-Rad) and chemiluminescence was quantified using Chemi-Doc (Bio-Rad).

2.4 | Surface plasmon resonance

The affinities of MBP-HRG constructs and synthetic HRR peptides for FXIIa, β -FXIIa, and FXII, and polyP were quantified using surface plasmon resonance (SPR) on a BIAcore T200. Streptavidin (ThermoFisher) was attached onto flow cells of a CM4 sensor chip to 2000–3000 response units (RU) using an amine coupling kit.²⁴ FXIIa or β -FXIIa, labeled at the active site with biotinylated Phe-Pro-Arg-chloromethyl ketone, was absorbed to immobilized streptavidin to ~ 500 RU at a flow rate of 10 μ l/min at 25°C. FXII was immobilized directly onto a flow cell of a CM4 chip to ~ 500 RU by amine coupling. Unmodified polyP was immobilized on a CM5 chip derivatized with a diamine linker. Ethylenediamine first was attached to the flow cell using an amine coupling protocol (Cytiva), where ethylenediamine was substituted for ethanolamine. PolyP was immediately attached to the free amine with modifications to the method of Choi et al.³⁸ and Thermo Scientific Tech Tip.³⁹ PolyP was dissolved in water to 500 mg/ml at 37°C. A solution consisting of 230 μ l of polyP, 50 μ l of 1 M EDC (in 100 mM 3-(N-morpholino)-propane sulfonic acid [MOPS], pH 6.5), 25 μ l of 0.1 M imidazole, pH 6.0, and 192 μ l of 100 mM MOPS, pH 6.5 was mixed and supplemented with an additional 25 μ l of 0.1 M imidazole. The solution was immediately injected into the flow cell at 5 μ l/min for 50 min, followed by two additional injections for 15 min each. This resulted in an increase in RU of 140. The running buffer was 100 mM MOPS, pH 6.5. The control flow cell was blocked with ethanolamine using amine coupling.

Binding experiments were performed at 25°C in HBS buffer containing 0.005% Tween-20 and 13 μ M ZnCl₂ at a flow rate of 50 μ l/min. Increasing concentrations of analytes, 0–500 nM HRG, 0–5000 nM HRG domain constructs, or 0–8000 nM synthetic HRR peptides, were injected into both flow cells for 300 s, followed by running buffer alone for 900 s. Flow cells were regenerated with 250 mM NaCl, 62.5 mM imidazole, and 5 mM EDTA for 60 s. Reciprocal experiments were performed by SPR to determine the affinity of FXII and FXIIa for HRG in the presence of 13 μ M ZnCl₂. Biotinylated HRG was absorbed to immobilized streptavidin onto a CM5 chip to 560 RU at a flow rate of 10 μ l/min at 25°C.²⁴ Binding experiments were performed as previously described with increasing

concentrations of 0–1000 nM FXII and FXIIa. Sensorgrams were analyzed with BIAcore T200 evaluation software Version 1.0 (BIAcore). Dissociation constant (K_d) values were determined from equilibrium values or calculated from rate constants using 1:1 Langmuir analysis by the instrument software.

2.5 | FXII autoactivation

In a 100- μ l reaction volume, silica (40 μ g/ml) or polyP (8 μ g/mL) was added to wells of a 96-well plate containing 6 μ M ZnCl_2 and 100 nM FXII in HBS-Tw20 in the absence or presence of MBP-HRG constructs or synthetic HRR peptides at concentrations up to 5000 nM. After 15 min incubation, 200 μ M S-2302 was added to each well and absorbance was monitored at 405 nm every 10 s for 30 min at 37°C using a Thermomax microplate reader (Molecular Devices, San Jose, CA). FXIIa generation was quantified by dividing rates of absorbance change by the specific activity of 100 nM FXIIa, which was determined separately. Plots of the concentrations of FXIIa generated versus the concentrations of HRG or HRG derivatives were subjected to nonlinear regression analysis using TableCurve 2D Version 4 (Jandel Scientific Software, SPSS Inc., Chicago, IL) and fitted to a rectangular hyperbola so that the concentration required for 50% inhibition of FXII activation (IC_{50}) could be calculated.

2.6 | FXI activation by FXIIa

In 100- μ l reaction volumes, HRG or synthetic HRR peptides (0–5000 nM) were added to wells containing 6 μ M ZnCl_2 , 2 nM FXIIa, 100 nM FXI, and 0.5 μ g/ml polyP in HBS-Tw20. After a 5-min incubation at 37°C, 1 μ M CTI was added to inhibit FXIIa. After an additional 5-min incubation, 200 μ M S-2366 was added and absorbance was monitored at 405 nm as described. FXIa generated was quantified by comparison with a standard curve of FXIa activity. The percentages of FXI activation were normalized relative to that obtained in the absence of HRG or synthetic HRR peptides and plotted against analyte concentration. IC_{50} values were determined as described previously.

2.7 | Prekallikrein activation by FXIIa

In 100- μ l reactions, HRG or synthetic HRR peptides (0–5000 nM) were added to wells of a 96-well plate containing 120 nM prekallikrein, 6 μ M ZnCl_2 , 5 μ g/ml polyP, and 2 mM CaCl_2 in 0.005% Tween20. To initiate the reactions, 2 nM FXIIa was added, and the mixtures were incubated for 10 min at 37°C before addition of 1 μ M CTI to inhibit FXIIa. After a 5-min incubation at the same temperature, 200 μ M S-2302 and 3 mM EDTA were added, and absorbance was measured at 405 nm for 60 min. Kallikrein generation was quantified by comparison with a standard curve of kallikrein activity. IC_{50} values were determined as described previously.

2.8 | Plasma clotting assays

In a 96-well plate, 90- μ l aliquots of control or HRG-depleted human plasma containing 0–5 μ M HRG or HRR peptides or 0–8 μ M MBP-HRG domain constructs were added to wells containing either 10 μ l of a 1:10 dilution of aPTT-SP or 10 μ l of a 1:100 dilution of RecombiPlasTin 2G reagent. After a 10-min incubation at 37°C, clotting was initiated by addition of 10 μ l of 260 mM CaCl_2 (final concentration 26 mM). Absorbance was monitored at 405 nm for 1 h at 37°C and the time to reach half-maximal absorbance as determined by instrument software was used as the clot time.

2.9 | Thrombin generation assay

Thrombin generation was quantified as previously described with some modifications.²⁵ HRG-depleted human plasma (50 μ l) containing 0–20 μ M HRG or HRR peptides was added to the wells of a black 96-well plate (Costar, Lowell, MA). To the wells was added 40 μ g/ml polyP and 15 μ M phosphatidylcholine-phosphatidylserine. After a 10-min incubation at 37°C, 15 mM CaCl_2 was added and thrombin generation was monitored by measuring the hydrolysis of 1 mM Z-Gly-Gly-Arg AMC (Bachem, Bubendorf, Switzerland) at excitation and emission wavelengths of 360 nm and 460 nm, respectively, for 90 min at 37°C using a SpectraMax M5e plate reader (Molecular Devices). Thrombin generation profiles and data were analyzed using Technothrombin TGA software (Technoclone, Vienna, Austria), as previously described.²⁵

2.10 | Statistical analysis

Results are presented as the mean $\pm$ SD. Significance of differences was determined using one or two-way analysis of variance followed by the Holm-Sidak comparison test. SigmaPlot (Systat Software, San Jose, CA) and IBM SPSS Statistics software (Armonk, NY) were used to create graphs and perform the statistical analyses, respectively. For all analyses, p values < .05 were considered statistically significant.

3 | RESULTS

3.1 | HRR and N1 domains of HRG bind FXIIa and polyP with high affinity

To identify the domains of HRG responsible for its interaction with FXIIa, individual domains were subcloned and expressed in *E coli* (Figure 1A). Recombinant MBP-HRG domain constructs (N1, N2, and N1N2; PRR1, PRR2; HRR; and COOH) were purified from *E coli* lysates using amylose affinity chromatography. SDS-PAGE (Figure 1B) and Western blot analysis (Figure 1C) revealed the expected insert sizes of the constructs.

The affinities of the MBP-HRG domain constructs for FXIIa, β -FXIIa, FXII, and polyP were determined by SPR and compared with the affinity of plasma-derived HRG. In the presence of Zn^{2+} , MBP-HRG domain constructs bind to immobilized FXIIa with varying affinities (Table 1), but do not bind to immobilized FXII or β -FXIIa. Like HRG, MBP-HRR binds to FXIIa with high affinity (K_d values of 22 ± 4 and 123 ± 7 nM, respectively). MBP-N1 and MBP-N1N2 also have similar affinities (K_d values of 92 ± 6 and 155 ± 26 nM, respectively). MBP-PRR2 binds to FXIIa with moderate affinity (K_d value of 307 ± 110 nM). In contrast, the remaining domains (MBP-N2, MBP-PRR1, MBP-COOH) bind FXIIa with significantly lower affinities than HRG. To exclude the possibility that the method of FXII or FXIIa immobilization could influence the binding data, reciprocal binding experiments

were performed with immobilized HRG. FXIIa binds HRG with a K_d value of 67 ± 0.9 nM (Figure S2), whereas FXII does not bind HRG (not shown).

Next, we determined the affinities of HRG and HRG domain constructs for polyP. Like HRG, MBP-HRR binds polyP with high affinity (K_d values of 115 ± 19 and 187 ± 19 nM, respectively) (Table 1). MBP-N1, MBP-N2, and MBP-N1N2 also bind polyP (K_d values of 71 ± 1 , 140 ± 1 , and 243 ± 1 , respectively). MBP-PRR1 and MBP-PRR2 bind polyP with lower affinity (K_d values of 949 ± 112 and 3400 ± 10 nM, respectively), whereas MBP control and MBP-COOH do not bind. Together, these findings suggest that, although the interaction of HRG with FXIIa and polyP may involve multiple domains, the HRR and N1 domains exhibit the highest affinity.

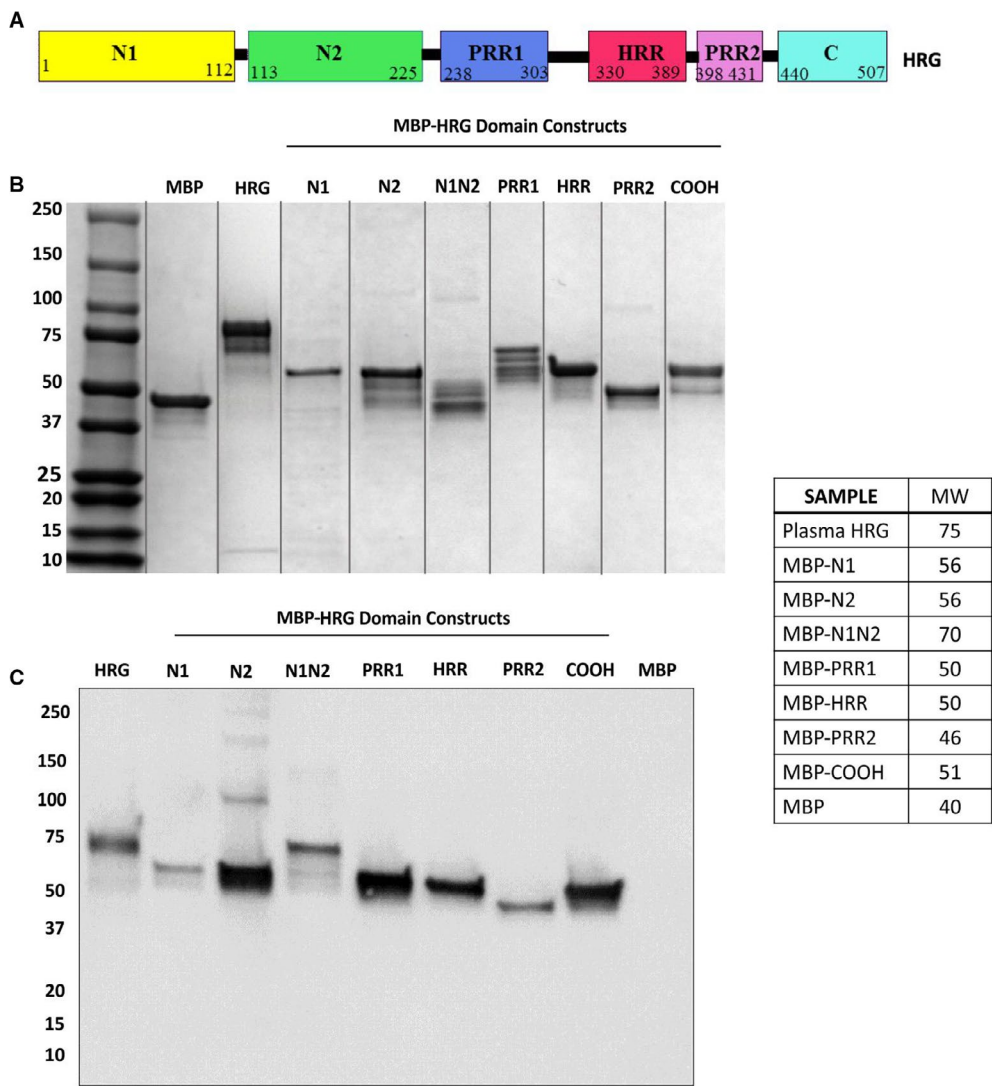


FIGURE 1 SDS-PAGE and Western blot analysis of MBP-HRG domain constructs. (A) Schematic diagram of HRG domain structure. MBP-HRG domain recombinant proteins and plasma HRG were subjected to SDS-PAGE analysis (B) on 4%–15% polyacrylamide gels in the presence of 0.1% sodium dodecyl sulfate (SDS) and 0.14 M β -mercaptoethanol. Images from separate gels have been spliced together in order from N-terminus. Separated proteins were transferred onto a nitrocellulose membrane and the membrane was probed with an affinity-purified polyclonal sheep IgG against HRG that was labeled with HRP (C). Molecular weights (kDa) of HRG and MBP-HRG domain constructs are provided in the right-hand table. C, carboxyl terminal domain (COOH); HRG, histidine-rich glycoprotein; HRR, histidine-rich region; N1/2, cystatin-like N1/N2 terminal domain; PRR1/2, proline-rich region; MBP, maltose-binding protein

TABLE 1 Dissociation constants for the interactions of HRG and MBP-HRG domain constructs with FXIIa and polyP

	FXIIa K_d (nM)	PolyP
Plasma HRG	22 ± 4	115 ± 19
MBP control	>5000	>5000
MBP-N1	92 ± 6	71 ± 1
MBP-N2	1037 ± 38	140 ± 1
MBP-N1N2	155 ± 26	243 ± 1
MBP-PRR1	4010 ± 163	949 ± 12
MBP-HRR	123 ± 7	187 ± 19
MBP-PRR2	307 ± 110	3400 ± 10
MBP-COOH	739 ± 33	>5000

Note: The affinities of plasma HRG and recombinant MBP-tagged HRG domain constructs for immobilized FXIIa and polyP were determined by surface plasmon resonance in the presence of 13 μ M ZnCl₂. K_d values represent the mean ± SD of three to four determinations.

Abbreviations: F, factor; HRG, histidine-rich glycoprotein; K_d , dissociation constant; MBP, maltose-binding protein; polyP, polyphosphate.

TABLE 2 Effect of recombinant MBP-HRG domain constructs on FXII autoactivation induced by polyP

	FXII autoactivation by polyP IC_{50} (nM)
Plasma HRG	275 ± 10
MBP control	>8000***
MBP-N1	3189 ± 275***
MBP-N2	1476 ± 420***
MBP-N1N2	809 ± 59*
MBP-PRR1	3840 ± 309***
MBP-HRR	509 ± 59
MBP-PRR2	652 ± 56
MBP-COOH	1519 ± 140***

Note: Data represent the mean ± SD of three experiments each done in duplicate * p < .05 and *** p < .001 compared with HRG (ANOVA, Holm-Sidak method).

Abbreviations: F, factor; HRG, histidine-rich glycoprotein; IC_{50} , 50% inhibition of FXII activation; MBP, maltose-binding protein; polyP, polyphosphate

3.2 | His/Pro-rich domains of HRG inhibit polyP-mediated FXII autoactivation

To examine the functional activity of the HRG domain constructs, their inhibitory effect on polyP-induced FXII autoactivation was examined and compared with that of plasma-derived HRG. Like HRG, MBP-HRG domain constructs inhibit FXII autoactivation in a concentration-dependent and saturable manner, whereas MBP control does not (Table 2). Plasma HRG inhibits FXII autoactivation by

over 85% with an IC_{50} of 275 ± 10 nM, whereas MBP-HRR and MBP-PRR2 inhibit FXII autoactivation with IC_{50} values of 509 ± 59 and 652 ± 56 nM, respectively. The high affinities of MBP-N1 and MBP-N1N2 for FXIIa do not endow them with potent inhibitory effects on FXII autoactivation (IC_{50} values 3189 ± 275 and 809 ± 59 nM, respectively). These findings suggest that even though the N1 domain binds to FXIIa, it is not an efficient inhibitor of FXII autoactivation. The remaining domains have significantly lower potencies compared with HRG (IC_{50} values >1400 nM). Together, these findings suggest that the HRR and PRR2 are the predominant domains of HRG responsible for inhibition of polyP-induced FXII autoactivation. We also examined the effect of MBP-HRG domain constructs on FXII autoactivation stimulated by silica, which we have previously shown does not bind HRG.²⁴ Like HRG, MBP-HRR inhibits FXII autoactivation stimulated by silica (IC_{50} 353 ± 149 and 143 ± 35, respectively; p < .001 vs HRG) (Table S3). MBP-PRR2 and MBP-COOH inhibit with lower potency (IC_{50} values of 1492 ± 11 and 967 ± 170 nM, respectively), whereas N-terminal domain constructs (MBP-N1, MBP-N2, MBP-N1N2) have no effect on FXII autoactivation (IC_{50} values >6000 nM). These results demonstrate that the inhibitory effect of HRG and derivatives is expressed in part through binding FXIIa.

3.3 | HRG prolongs the aPTT

We have previously shown that addition of HRG to plasma prolongs the APTT but has little effect on the PT.²⁴ This confirms that the anticoagulant effect of HRG is expressed through FXII because the APTT reagent is silica-based. In the current study, MBP-HRR prolongs the APTT in HRG-depleted plasma by 7.2-fold, comparable with the 8.8-fold prolongation with HRG (Figure 2A). MBP-PRR2 also prolongs the APTT by 2.3-fold, whereas the remaining constructs (MBP-N1, MBP-N2, MBP-N1N2, MBP-PRR1, and MBP-COOH) produce less than two-fold prolongation. HRG and the HRG domain constructs produce less than two-fold prolongation of the PT (Figure 2B). These results support the concept that the effect of HRG is mainly localized to the intrinsic pathway of coagulation, and that the anticoagulant activity of HRG is predominantly mediated by the HRR.

3.4 | HRR peptides bind FXIIa and polyP with high affinity

Our current findings suggest that the HRG-FXIIa interaction is mediated by the HRR domain of HRG. To further characterize the features of HRR that enable its regulation of FXIIa activity, we synthesized HRR peptide analogs containing one or four repeats of the consensus sequence: (HHPHG)₁ and (HHPHG)₄, respectively.³⁶ A related scrambled peptide, Scr(HAHG)₅H, containing the same number of His residues, and poly-His peptides consisting of six or 12 His residues (H-6 and H-12, respectively) (Table S2). The affinities of HRR peptides for FXIIa, β -FXIIa, FXII, and polyP were determined

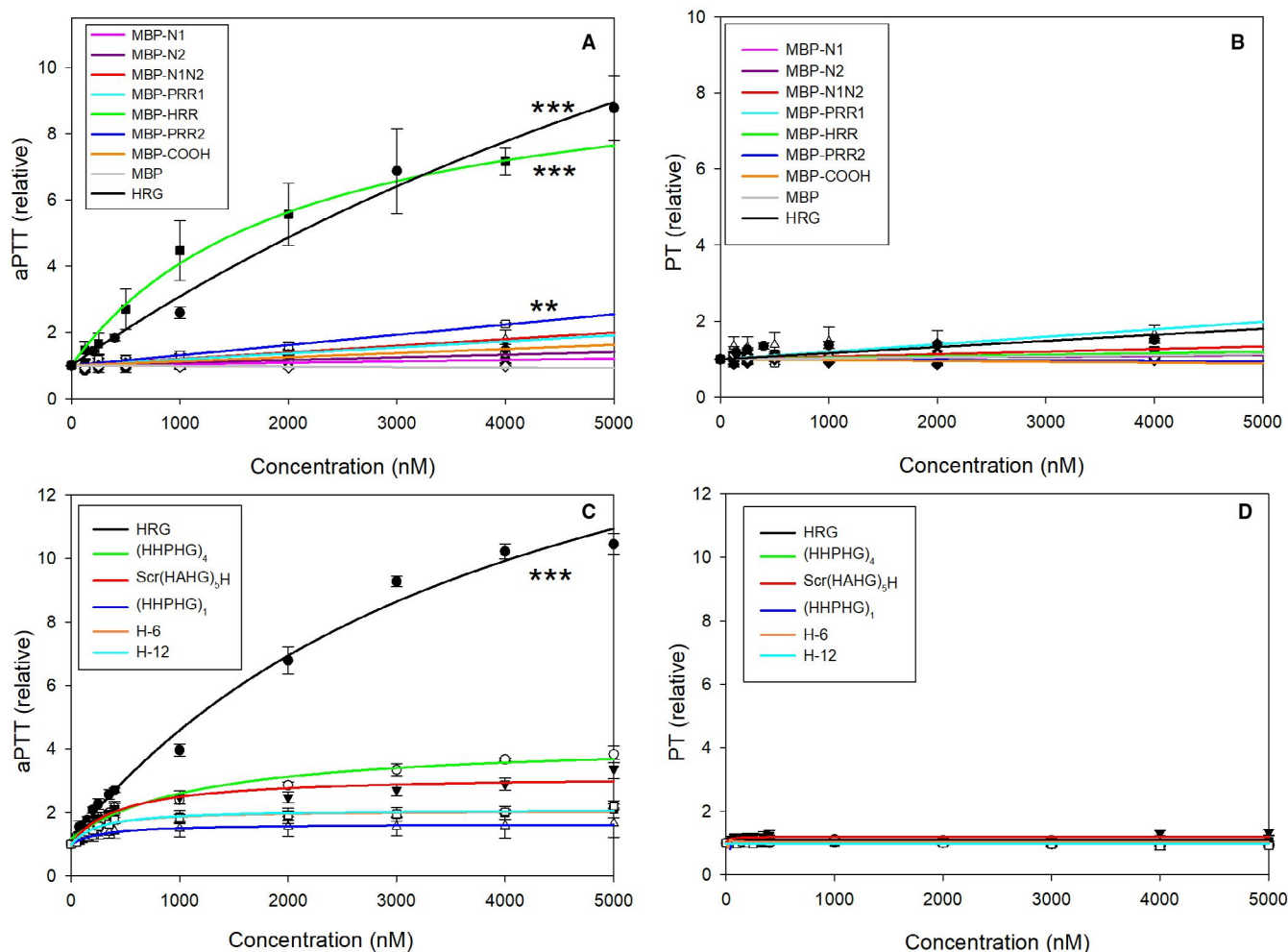


FIGURE 2 Effect of HRG, recombinant MBP-HRG domain constructs, and HRR peptides on the APTT and PT in HRG-depleted human plasma. HRG (0–5000 nM), MBP-HRG domain constructs (0–4000 nM), or HRR peptides (0–5000 nM) was added to HRG-depleted plasma and incubated with 1:10 APTT-SP reagent (A, C) or 1:100 RecombiPlasTin reagent (B, D) for 10 min at 37°C before initiating clotting with 26 mM CaCl_2 . The APTT or PT was determined and normalized relative to that measured in the absence of HRG, MBP-HRG domain constructs, or HRR peptides. Data points represent the mean of three experiments each done in duplicate; bars reflect SD. Data were fitted by nonlinear regression analysis (lines). ** $p < 0.01$ and *** $p < 0.001$ compared with control (ANOVA, Holm-Sidak method). APTT, activated partial thromboplastin time; HRG, histidine-rich glycoprotein; HRR, histidine-rich region; MBP, maltose-binding protein; PT, prothrombin time

by SPR and compared with the affinity of plasma-derived HRG. In the presence of Zn^{2+} , HRG and $(\text{HHPHG})_4$ bind to immobilized FXIIa with K_d values of 12 ± 0.03 nM and 222 ± 20 nM, respectively, whereas $\text{Scr}(\text{HAHG})_5\text{H}$ binds with a K_d value of 443 ± 0.9 nM (Table 3). Binding of HRG and HRR peptides to FXIIa is Zn^{2+} dependent because binding is abrogated with imidazole, a His analog. The monomer peptide, $(\text{HHPHG})_1$ does not bind to FXIIa nor do H-6 and H-12. Therefore, His residues alone are insufficient to support FXIIa binding. Unlike plasma HRG, HRR peptides do not bind FXIIa in the absence of Zn^{2+} (not shown). Neither HRG nor HRR peptides bind to immobilized β -FXIIa or FXII in the absence or presence of Zn^{2+} . For polyP, plasma-derived HRG binds with a K_d value of 193 ± 11 nM in the presence of $13 \mu\text{M}$ ZnCl_2 (Table 3). $(\text{HHPHG})_4$, $\text{Scr}(\text{HAHG})_5\text{H}$, and H-6 bind polyP with lower affinity (K_d values of 1067 ± 54 , 1376 ± 268 , and 2750 ± 250 nM, respectively), whereas $(\text{HHPHG})_1$

and H-12 do not bind (K_d values >8000 nM). Together, these findings support our hypothesis that the binding of HRG to FXIIa and polyP is primarily mediated by the HRR.

3.5 | HRR peptides inhibit FXII autoactivation

Given that HRR peptides bind FXIIa and polyP with high affinity, we hypothesized that like HRG, synthetic HRR-derived peptides would exhibit anticoagulant properties. Compared with HRG (IC_{50} of ~ 200 nM), HRR peptides more potently inhibit polyP-induced FXII autoactivation (Table 4). Specifically, $(\text{HHPHG})_4$ inhibits FXII autoactivation with an IC_{50} 59 ± 18 nM, whereas $\text{Scr}(\text{HAHG})_5\text{H}$ and H-12 inhibit FXII autoactivation with IC_{50} values of 146 ± 14 and 114 ± 32 nM, respectively. In contrast, $(\text{HHPHG})_1$ has no effect on

FXII autoactivation. Thus, like HRG, HRR peptides containing more than 11 His residues inhibit polyP-induced FXII autoactivation.

3.6 | HRR peptides inhibit FXIIa-mediated activation of FXI and prekallikrein

The effect of HRG and HRR peptides on FXI activation by FXIIa was assessed in the presence of ZnCl_2 and polyP (Table 4). Like HRG, HRR peptides inhibit FXI activation by FXIIa in the presence of polyP and ZnCl_2 . (HHPHG)₄, Scr(HAHG)₅H, and H-12 inhibit FXI activation by more than 90% (IC_{50} values of 337 ± 50 , 256 ± 28 , and 634 ± 65 , respectively). HRG is significantly more potent than HRR peptides with an IC_{50} of 51 ± 7 ($p < .001$). (HHPHG)₁ has no effect on FXI activation. These results show that, like HRG, HRR peptides attenuate FXIIa-mediated activation of FXI. Next, the effect of HRG or

HRR peptides on FXIIa-mediated activation of PK was assessed in the presence of Zn^{2+} and polyP (Table 4). Like HRG, HRR peptides inhibit PK activation by 30% to 60%. However, HRG is significantly more potent than the HRR peptides ($p < .001$) (Table 4). (HHPHG)₁ inhibits by only 18% with an IC_{50} value >5000 nM. These findings suggest that HRR peptides can mimic the inhibitory effect of HRG on FXIIa-mediated activation of PK.

3.7 | HRR peptides prolong clotting in human plasma via the intrinsic pathway

Addition of HRG or HRR peptides prolongs the APTT in HRG-depleted plasma. HRG significantly prolongs the APTT by 10.5-fold, whereas (HHPHG)₄ and Scr(HAHG)₅H prolong the aPTT by 3.8-fold and 3.4-fold, respectively ($p < .001$ compared with control) (Figure 2C). H-6 and H-12 prolong the APTT by ~two-fold, which is similar to the effect of the monomer peptide (1.7-fold). HRG and HRR peptides have little to no effect on the PT (<1.3 -fold prolongation) (Figure 2D). Together, these results confirm that the effect of HRG is localized to the intrinsic pathway of coagulation and demonstrate the important role of the HRR domain in modulating the anticoagulant properties of HRG.

3.8 | HRR peptides attenuate polyP-induced thrombin generation

Finally, the effect of HRG and HRR peptides on thrombin generation induced by 40 $\mu\text{g}/\text{ml}$ polyP was examined in HRG-depleted human plasma. This concentration of polyP was chosen because pilot experiments revealed that it induced thrombin generation in HRG-depleted human plasma (not shown). HRG and HRR peptides significantly prolong the time to peak thrombin and decrease peak thrombin in a concentration-dependent manner (Figure 3A–D).

TABLE 3 Dissociation constants for the interactions of HRG and synthetic HRR peptides with FXIIa and polyP

	FXIIa Kd (nM)	PolyP
HRG	12 ± 0.03	193 ± 11
(HHPHG) ₄	222 ± 20	1067 ± 54
Scr(HAHG) ₅ H	443 ± 0.9	1376 ± 268
(HHPHG) ₁	>5000	>8000
H-12	>5000	>8000
H-6	>5000	2750 ± 250

Note: The affinities of plasma HRG and synthetic HRR peptides for immobilized FXIIa or polyP were determined by surface plasmon resonance. Responses that did not show saturation are represented by the highest concentration tested. Dissociation constant (K_d) values represent the mean $\pm$ SD of three to four determinations.

Abbreviations: F, factor; HRG, histidine-rich glycoprotein; HRR, histidine-rich region; polyP, polyphosphate.

	FXII autoactivation by polyP IC_{50} (nM)	FXI activation by FXIIa	PK activation by FXIIa
HRG	275 ± 10	476 ± 14	163 ± 25
(HHPHG) ₄	$59 \pm 18^{***}$	$225 \pm 21^{***}$	$512 \pm 13^{***}$
Scr(HAHG) ₅ H	$146 \pm 14^{***}$	$392 \pm 18^{***}$	$343 \pm 21^{***}$
H-12	$114 \pm 32^{***}$	$781 \pm 16^{***}$	$797 \pm 29^{***}$
H-6	270 ± 10	$824 \pm 24^{***}$	$1677 \pm 38^{***}$
(HHPHG) ₁	$>5000^{***}$	$>5000^{***}$	$>5000^{***}$

Note: FXII autoactivation was induced with 8 $\mu\text{g}/\text{ml}$ polyP. Effects of HRG or HRR peptides on activation of FXI by FXIIa were assessed in the presence of 0.5 $\mu\text{g}/\text{ml}$ polyphosphate. Activation of 120 nM PK by FXIIa was determined in the presence of 5 $\mu\text{g}/\text{ml}$ polyP. IC_{50} values were determined by nonlinear regression analysis. Data represent the mean $\pm$ SD of three experiments each done in duplicate. $^{***}p < .001$ compared with HRG (ANOVA, Holm-Sidak method).

Abbreviations: F, factor; HRG, histidine-rich glycoprotein; HRR, histidine-rich region; IC_{50} , 50% inhibition of FXII activation; polyP, polyphosphate.

TABLE 4 Effect of HRG and HRR peptides on FXII autoactivation and FXIIa-mediated reactions

HRG and Scr(HAHG)₅H prolong the time to peak thrombin by 2.5-fold and two-fold, respectively (Figure 3C), whereas (HHPHG)₄ and H-12 prolong it by 3.1-fold and 3.4-fold, respectively. HRG and HRR peptides decrease peak thrombin up to ~2.5-fold in a concentration-dependent and saturable manner (Figure 3D), whereas (HHPHG)₁ has little effect. These findings suggest that, like HRG, HRR peptides have the capacity to downregulate the procoagulant effect of polyP in human plasma.

4 | DISCUSSION

Previously, we showed that HRG serves as a molecular brake for the intrinsic pathway of coagulation by binding FXIIa and limiting its interaction with FXII and FXI and by binding polyP and attenuating its capacity to promote FXII autoactivation.^{24,34,35} We have also shown that intraperitoneal polyP induces thrombosis in mice in a FXII-dependent fashion, confirming the prothrombotic properties of polyP.³⁵ In the current study, we demonstrate that (1) the interaction of HRG with FXIIa and polyP is predominantly mediated by the HRR domain and (2) HRR analogs inhibit

polyP-induced FXII autoactivation and FXIIa-mediated activation of FXI and PK, prolong the APTT, and attenuate polyP-induced thrombin generation in HRG-depleted human plasma. Therefore, these findings identify the HRR domain as the primary mediator of HRG's effect on the contact system.

Studies with individually expressed recombinant HRG domains reveal that compared with intact HRG, HRR has comparable affinity for FXIIa and produces similar prolongation of the APTT. The HRR of human HRG contains 12 tandem repeats of the five-residue Gly-His-His-Pro-His consensus sequence.^{29,30} By examining His-rich peptides based on this sequence, we show that the anticoagulant properties of HRG are predominantly mediated by the abundant His residues in the HRR domain. The peptides used in these studies include a tetramer of the consensus sequence, (HHPHG)₄, as well as a scrambled version and poly-His H-6 and H-12. Of these, the tetramer peptide has the highest potency for inhibiting FXIIa and attenuates thrombin generation to a similar extent as HRG. These results suggest that although His residues are important for the interaction of HRG with FXIIa, binding also is sequence dependent. To better understand the role of His residues in the interaction between HRG and FXIIa, we look to HK,

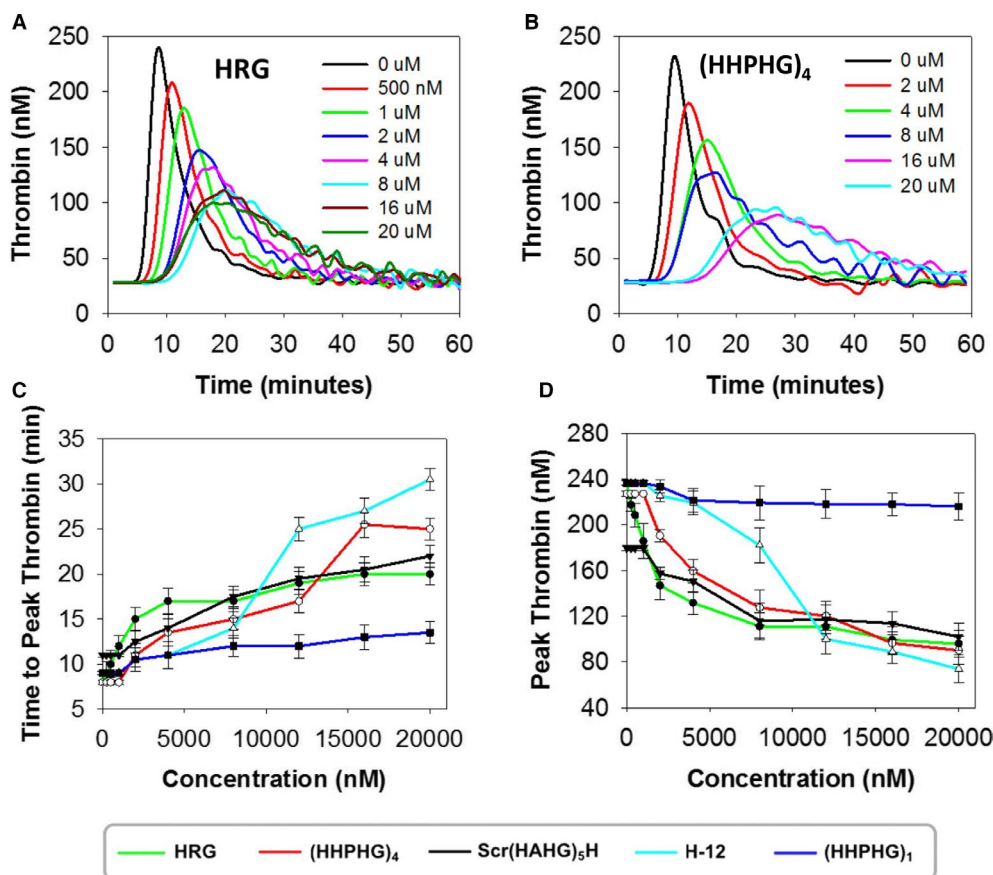


FIGURE 3 Effect of HRG and HRR peptides on polyphosphate-induced thrombin generation in HRG-depleted human plasma. Thrombin generation in HRG-depleted citrated human plasma was triggered with 40 μ g/ml polyP and 15 mM CaCl_2 in the presence of 0–20 μ M HRG (A) or (HHPHG)₄ (B) and monitored by measuring the hydrolysis of Z-Gly-Gly-Arg-AMC. Time to peak thrombin (C) and peak thrombin concentration (D) are plotted. Representative data are provided in panels A and B; data points in panels C and D represent the mean of three experiments each done in duplicate; the bars reflect the SD. HRG, histidine-rich glycoprotein; HRR, histidine-rich region; polyP, polyphosphate

which shares homology with HRG because the D5_H region in the light chain of HK contains abundant histidine residues.⁴⁰ It is this region of HK that mediates its interaction with anionic surfaces, Zn²⁺, and heparin.^{40,41} However, HK does not bind to FXIIa, suggesting that the HRG interaction requires more than His residues in the HRR.⁴² Therefore, despite the abundance of His residues in HK, HRG exhibits a unique His-dependent interaction with FXIIa suggesting that the interaction of HRG with FXIIa is dependent on both sequence and conformation.

HRG has the capacity to affect the contact system in two ways. This occurs because HRG binds and antagonizes both FXIIa and polyanions such as polyP. HRG, recombinant HRG domains, and HRR analogs inhibit silica-induced FXII autoactivation and prolong clotting in silica-based APTT assay in HRG-depleted human plasma. These anticoagulant effects likely reflect binding to FXIIa because HRG does not bind silica.²⁴ Furthermore, HRG has previously been shown to attenuate thrombin-mediated activation of FXI in the presence of RNA.³⁴ Thus, HRG also can modulate coagulation without binding FXIIa.

The site of interaction of HRG on FXIIa is uncertain. HRG likely binds to the heavy chain of FXIIa because HRG does not affect the rate of FXIIa inhibition by active site-directed macromolecular inhibitors such as CTI or C1-INH, and HRG does not bind the catalytic light chain of FXIIa.²⁴ Because the interaction of HRG and FXIIa is Zn²⁺ dependent, it is possible that a conformational change in HRG is necessary for interaction, as observed for other HRG interactions.^{26,33,42} Previous work showed that the interaction of HRG with FXIIa may also involve the heparin binding site within the N1 or N2 domains.²⁴ Binding of heparin to the NH₂-terminal cystatin domain of HRG is enhanced when Zn²⁺ binds to the HRR domain.^{27,29,31,32} Our data showing that the N1 and N1N2 domain constructs exhibit high affinity for FXIIa support this possibility. Thus, it is possible that both the N1 and the HRR domains mediate the interaction of HRG with FXIIa, and that Zn²⁺ may directly and indirectly regulate this interaction. These results add to the multifunctional roles of the HRG domains^{36,43–46} and provide further insight into the mechanism by which HRG modulates the contact system.^{24,25,34,47}

In addition to its high levels in plasma, HRG is released from activated platelets,⁴⁸ and we have shown that platelets and HRG colocalize in thrombi induced in mice with FeCl₃²⁵ or polyP.³⁵ Therefore, HRG is well situated to regulate the contact system at sites of vascular injury, whether induced by nucleic acids in neutrophil extracellular traps or by polyP released from platelets. Accordingly, HRG attenuates thrombosis induced in mice either with polyP or FeCl₃.^{25,35} Moreover, HRG is a negative acute phase protein whose levels decrease in pregnancy or with infection, inflammation, or advanced cancer, conditions associated with an increased risk for thrombosis.^{27,49–52} Identification of the domains responsible for the anticoagulant role of HRG may provide opportunities to supplement HRG activity to prevent or treat thrombosis. Thus, identification of HRR as the domain responsible for the ability of HRG to downregulate the contact system may provide novel strategies to attenuate thrombosis without disruption of hemostasis.

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CONFLICT OF INTEREST

The other authors declare no competing financial interests.

AUTHOR CONTRIBUTIONS

T.K. Truong designed research, contributed vital new reagents, performed research, analyzed and interpreted data, and wrote the manuscript; A.R. Stafford designed research, contributed vital new reagents, and analyzed and interpreted data; R.A. Malik designed research, contributed vital new reagents, performed research, and analyzed and interpreted data; X. Yao and H.M. Madarati designed and performed research, and analyzed and interpreted data; C.A. Kretz designed research, and analyzed and interpreted data; J.C. Fredenburgh designed research, contributed vital new reagents, performed research, analyzed and interpreted data, and wrote the manuscript; and J.I. Weitz designed research, analyzed and interpreted data, and wrote the manuscript.

ORCID

Tammy K. Truong  <https://orcid.org/0000-0003-0936-6757>

Rida A. Malik  <https://orcid.org/0000-0001-7118-8715>

Xintong Yao  <https://orcid.org/0000-0002-3627-0331>

James C. Fredenburgh  <https://orcid.org/0000-0003-1223-0754>

Hasam M. Madarati  <https://orcid.org/0000-0003-0983-678X>

Colin A. Kretz  <https://orcid.org/0000-0001-7979-7603>

Jeffrey I. Weitz  <https://orcid.org/0000-0002-1092-7550>

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SUPPORTING INFORMATION

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